

UI/UX Specification: AI & LMS-Enabled Cooperative Capacity Building, ERP, and Employment Ecosystem

1. Design System Architecture ("Sahkar Design System - SDS")

The Sahkar Design System (SDS) provides a unified, enterprise-grade design framework tailored for the National Council for Cooperative Training (NCCT), its constituent institutes (VAMNICOM, RICMs, ICMs), and the broader cooperative ecosystem under the Ministry of Cooperation. It balances the institutional authority of a central government platform with an accessible, high-performance interface suited for rural learners and cooperative staff.

Design Philosophy and Core Tenets

- **Radical Clarity & Low Cognitive Overhead:** Complex enterprise workflows (such as batch scheduling, hostel allocation, and multi-state nominations) are divided into structured, progressive steps.
- **Dignity, Inclusion & Indic-First Experience:** Designed to empower rural learners, PACS secretaries, and women in Self-Help Groups (SHGs) through voice-assisted interfaces, intuitive visual affordances, and full vernacular localization.
- **Absolute Reliability in Variable Connectivity:** Interfaces adapt dynamically to network quality, providing clear offline states, persistent local storage, and seamless background synchronization.
- **Perspective Abstraction:** Low-level engineering mechanisms (e.g., cryptographic keys, biometric embeddings, and vector similarity models) are abstracted into clear, friendly user interactions.

Design Token Architecture

Spacing Tokens

The system uses a strict 4px baseline grid to maintain visual rhythm across form factors:

- space-2xs: 2px (tight icon-text offsets)
- space-xs: 4px (input padding adjustments)
- space-sm: 8px (component internal padding, button gutters)
- space-md: 16px (standard card padding, form field gaps)
- space-lg: 24px (card section spacing, table row gutters)
- space-xl: 32px (page layout gutters, dashboard widget separations)
- space-2xl: 48px (major page section dividers)
- space-3xl: 64px (hero canvas margins)

Border Radius Tokens

- radius-none: 0px (edge-to-edge full-bleed mobile banners)
- radius-sm: 4px (data tables, code snippets, micro-tags)

- radius-md: 8px (form inputs, regular buttons, standard metric cards)
- radius-lg: 12px (course overview cards, dialogue boxes, modal panels)
- radius-xl: 16px (feature banners, primary bottom sheets on mobile)
- radius-full: 9999px (circular status badges, voice-command buttons, avatar chips)

Elevation and Shadows

- elevation-0: Flat surface, border: 1px solid var(--border-subtle)
- elevation-1: Subtle lift for cards and table rows on hover: 0 1px 3px rgba(16, 24, 40, 0.08), 0 1px 2px rgba(16, 24, 40, 0.04)
- elevation-2: Dropdown menus, floating search bars, popover menus: 0 4px 8px -2px rgba(16, 24, 40, 0.08), 0 2px 4px -2px rgba(16, 24, 40, 0.04)
- elevation-3: Modals, dynamic QR attendance dialogs, slide-over panels: 0 12px 20px -4px rgba(16, 24, 40, 0.12), 0 4px 6px -2px rgba(16, 24, 40, 0.05)
- elevation-4: Persistent floating actions and Indic voice assistant drawer: 0 20px 28px -6px rgba(16, 24, 40, 0.16), 0 8px 10px -4px rgba(16, 24, 40, 0.08)

2. Color Palette, Gradients, and Visual Foundations

The color system combines colors representing the cooperative and agricultural sectors with enterprise tones that project institutional trust, meeting WCAG 2.1 AA/AAA contrast ratios across all screen types.

Color Tokens

Semantic Role	Token Name	Light Mode Value	Dark Mode Value	Practical Interface Context
Primary Brand (Coop Emerald)	--color-primary-600	#0E765E	#16A37F	Primary action buttons, active navigation states, completed progress tracks.
Primary Hover	--color-primary-700	#0A5C49	#22C55E	Interactive hover and pressed states for primary buttons.
Primary Light / Surface	--color-primary-50	#E8F5F1	#OD332A	Badge backgrounds, selected card

				outlines, active tab indicators.
Institutional Navy	--color-navy-900	#0F172A	#F8FAFC	Main application titles, navigation bar background, high-priority numbers.
Institutional Slate	--color-slate-600	#475569	#94A3B8	Secondary labels, metadata strings, inactive tab text, table headers.
Saffron / Warm Accent	--color-accent-500	#E67E22	#F39C12	DigiLocker certification highlights, pending tasks, action reminders.
Canvas Background	--color-bg-base	#F8FAFC	#090D16	Main window surface, page body canvas.
Surface Card	--color-bg-surface	#FFFFFF	#131B2E	Container cards, data tables, modal dialog panels.
Border Subtle	--color-border-subtle	#E2E8F0	#1E293B	Card dividers, input container boundaries, table row separators.
Semantic	--color-succes	#10B981	#34D399	Attendance verified,

Success	s-500			course passed, credential verified, sync complete.
Semantic Warning	--color-warnin g-500	#F59E0B	#FBBF24	Low attendance warning (<80%), sync pending, expiring deadlines.
Semantic Destructive	--color-error-500	#EF4444	#F87171	Duplicate nomination, failed assessment, room overbooked, invalid scan.
Semantic Info / Bridge	--color-info-500	#2563EB	#60A5FA	Government scheme links, DigiLocker sync status, career guidance alerts.

Professional Gradients

- Hero Gradient (Institutional Banner):**
 linear-gradient(135deg, #0A5C49 0%, #0E765E 55%, #1B4D3E 100%)
 Used for executive summary banners, national outreach metric panels, and certificate presentation covers.
- Surface Card Accent Gradient (Delight Accent):**
 linear-gradient(180deg, rgba(14, 118, 94, 0.04) 0%, rgba(255, 255, 255, 0) 100%)
 Applied across candidate summary cards, job matching widgets, and module progress containers to add subtle depth.
- DigiLocker Certified Glow:** linear-gradient(90deg, #F97316 0%, #EA580C 100%)
 Applied to verified badges, credential verification headers, and official Ministry certification seals.
- Offline Resilient Mode Banner:** linear-gradient(90deg, #334155 0%, #1E293B 100%)
 Subtle top bar indicating active offline caching and local record protection without

alarming the user.

3. Modern Typography & Multilingual Hierarchy

The platform uses a font pairing strategy that combines modern sans-serif fonts for Latin alphanumeric characters with Google's Noto Sans Indic font family to support 22 Scheduled Indian Languages.

- **Primary Latin & Numerals:** *Plus Jakarta Sans* (alternatively *Inter* for standard enterprise builds)
- **Indic Scripts (Multilingual Engine):** *Noto Sans Devanagari*, *Noto Sans Tamil*, *Noto Sans Telugu*, *Noto Sans Bengali*, *Noto Sans Gujarati*, etc.

Typographic Scale

Style Token	Font Size	Line Height	Font Weight	Letter Spacing	Typical Application
Display 1	36px (2.25rem)	44px (1.22)	700 Bold	-0.02em	Executive KPI metrics, public portal landing hero titles.
Display 2	30px (1.875rem)	38px (1.26)	700 Bold	-0.01em	Institute dashboard titles (VAMNICOM, RICM, ICM).
Heading 1 (H1)	24px (1.5rem)	32px (1.33)	600 Semi-Bold	-0.01em	Page headers, module titles, course titles in LMS.
Heading 2 (H2)	20px (1.25rem)	28px (1.40)	600 Semi-Bold	0.00em	Card grouping titles, hostel block

					headers, batch IDs.
Heading 3 (H3)	16px (1.0rem)	24px (1.50)	600 Semi-Bold	0.00em	Modal titles, table column groups, candidate profile names.
Body Large	16px (1.0rem)	26px (1.62)	400 Regular	0.00em	Conversatio nal AI responses, course description s, news feeds.
Body Regular	14px (0.875rem)	22px (1.57)	400 Regular	0.00em	Standard ERP table cells, input forms, body description s.
Body Medium	14px (0.875rem)	22px (1.57)	500 Medium	0.00em	Interactive buttons, navigation menu labels, filter chips.
Caption / Small	12px (0.75rem)	16px (1.33)	500 Medium	+0.01em	Timestamp s, sync status labels, field helper text, QR timers.
Overline / Tag	11px (0.6875rem)	14px (1.27)	700 Bold	+0.05em	Micro-badg es (VERIFIED,

	)				OFFLINE, GOVT ACCREDITE D).
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4. Responsive Breakpoints & Device Adaptability

The system uses a fluid layout engine designed to scale across desktop workstations, lecture-hall tablets, and affordable Android smartphones commonly used in rural areas.

Breakpoint Name	Viewport Width	Grid Columns	Gutters & Margins	Primary Device Focus & Adaptive UI Behavior
Compact (Mobile)	< 640px	4 Columns	16px Gutters / 16px Margins	Android smartphones; bottom navigation bar; full-width stacked cards; sticky bottom CTA sheets.
Medium (Tablet)	640px - 1023px	8 Columns	20px Gutters / 24px Margins	Tablets and foldable screens; collapsible navigation rail; 2-column card layouts; split-pane drawers.
Expanded (Desktop)	1024px - 1439px	12 Columns	24px Gutters / 32px Margins	Institutional desktops and laptops; persistent expandable sidebar; multi-column ERP tables;

				floating utility drawers.
Wide (Command Center)	≥ 1440px	12 Columns	32px Gutters / 48px Margins	High-res monitors; comprehensive MIS dashboards; multi-pane hostel grid and batch scheduling maps.

Adaptive Navigation Architecture

- **Mobile Experience (< 640px):**
 - *Bottom Navigation Bar:* 4 core destinations (Home, My Learning, Attendance, Career) with an elevated center Floating Action Button (FAB) for the Indic Voice Assistant ("सहकार वाणी").
 - *Top App Bar:* Institute logo, offline sync pill, language switcher dropdown, and notification bell.
 - *Contextual Drawers:* Sub-actions and filters appear as swipeable bottom sheets with thumb-accessible dismiss buttons.
- **Tablet Experience (640px - 1023px):**
 - *Collapsible Navigation Rail:* 72px slim vertical rail on the left displaying icons, expanding to 240px upon tap.
 - *Split-Pane View:* Master-detail browsing (e.g., list of training batches on the left, batch details and timetable on the right).
- **Desktop / Web ERP Experience (≥ 1024px):**
 - *Persistent Dual-Tier Sidebar:* Left-hand 260px navigation sidebar with collapsible nested modules (Academic ERP, LMS, Attendance, Logistics, Employment, Analytics).
 - *Universal Command Bar (Ctrl + K / Cmd + K):* Quick-jump global search across trainees, courses, batch IDs, hostel rooms, and job postings.
 - *Top Status Strip:* Trainee profile card, network status indicator, tenant institute switcher (for Ministry admins), and accessibility font-scaling tools.

5. Role-Specific User Perspectives & Interface

Abstraction

The interface design is structured around distinct user roles, presenting relevant controls while abstracting complex background operations into clear, intuitive interactions.

1. Trainee / Rural Youth Perspective (Mobile & PWA-First)

- **Goal:** Discover training opportunities, complete learning modules, track attendance,

obtain verified certificates, and find employment.

- **Key Interface Elements:**

- *Daily Routine Card*: Highlights today's training sessions, scheduled room numbers, and meal times.
- *Quick Check-in Button*: A prominent primary button on the home screen that opens the camera to scan the dynamic session QR code or activate edge face check-in.
- *Learning Hub*: Downloadable micro-lessons with progress bars, visual completion badges, and an audio listen button.
- *My Credentials*: Displays digital certificates featuring the official Ministry of Cooperation seal, with a single button to "Save to DigiLocker".

- **Behind-the-Scenes Abstraction:**

- *Edge Facial Recognition*: Trainees see an intuitive oval viewfinder that displays "Align face here" and transitions to a checkmark: "You're checked in for Lecture Hall 3" (avoiding technical references to 512D ArcFace embeddings or cosine thresholds).
- *Offline Sync*: A status indicator displays "Working Offline • 6 lessons ready" and updates to "Synced" when online (masking CRDT delta synchronizations).

2. Faculty / Trainer Perspective (Tablet & Desktop-First)

- **Goal**: Conduct training sessions, present dynamic attendance codes, monitor learner comprehension, and grade assessments.

- **Key Interface Elements:**

- *Session Console*: A clean interface displaying the active batch name, current attendance count (e.g., "38 of 42 present"), and session countdown timer.
- *Dynamic QR Display Mode*: Renders an updated session QR code with a circular countdown ring indicating the 15-second refresh cycle.
- *Student Progress Grid*: Displays learner completion cards, highlighting trainees who may need academic support.

- **Behind-the-Scenes Abstraction:**

- The cryptographic signing of dynamic QR payloads occurs seamlessly in the background without requiring instructor configuration.

3. Institutional Training Coordinator (RICM / ICM Desktop ERP)

- **Goal**: Manage course schedules, review society nominations, allocate classrooms and hostel rooms, and audit attendance logs.

- **Key Interface Elements:**

- *Batch Scheduling Matrix*: A visual drag-and-drop calendar that highlights resource conflicts across classrooms, computer labs, and instructors.
- *Nomination Approval Hub*: A candidate management view featuring bulk actions to review, verify, and approve applications from PACS and dairy federations.
- *Hostel Inventory Grid*: An interactive layout of residential dormitories displaying room availability, maintenance states, and current occupants.

- **Behind-the-Scenes Abstraction:**

- The linear optimization algorithm for room allocation runs behind a single "Auto-Assign Rooms" button, generating optimized assignments in seconds while respecting gender

segregation and cohort rules.

4. Cooperative Employer / Recruiter (Desktop & Tablet)

- **Goal:** Find qualified, certified personnel for positions in PACS, cooperative banks, and dairy processing societies.
- **Key Interface Elements:**
 - *Talent Discovery Engine:* Search candidates by district, certified skills, and assessment grades.
 - *Candidate Card:* Highlights verified skills, completed NCCT institutes, assessment scores, and a "Verified by Ministry of Cooperation" badge.
 - *Recruitment Pipeline:* A kanban board with columns for Applied, Shortlisted, Interview Scheduled, and Hired.
- **Behind-the-Scenes Abstraction:**
 - Vector similarity calculations and geographic distance-decay algorithms are distilled into a clear "Match Score" (e.g., "94% Match • Located 12 km from Society") with understandable match reasons.

5. Apex Leadership & Ministry Dashboard (Command Center)

- **Goal:** Monitor national training targets, review state-level PACS computerization progress, and evaluate employment outcomes.
- **Key Interface Elements:**
 - *National Capacity Heatmap:* An interactive map of India showing training volumes, seat utilization, and attendance rates across VAMNICOM, RICMs, and ICMs.
 - *Strategic Program Trackers:* Dedicated metric cards tracking initiatives like the PACS Computerization Training Mission and New Multipurpose Societies.
 - *Employment Linkage Analytics:* Real-time indicators showing candidate placements and recruitment activity across cooperative federations.

6. Cognitive Load Reduction & Standard UX Heuristics

To support users with varying levels of digital literacy across rural and institutional settings, the interface incorporates core human-computer interaction principles.

Application of UX Laws

- **Hick's Law (Minimizing Decision Time):**
 - Course nomination workflows are broken into 3 clear steps: Candidate Details, Sponsoring Society Selection, and Document Confirmation.
 - Trainee home screens present a single primary action at any given time (e.g., "Mark Today's Attendance" during class hours, or "Continue Module 2" during study periods).
- **Miller's Law (Information Chunking):**
 - Trainee profiles organize information into clear, bite-sized sections: Identity, Academic History, Skills, and Verifications.
 - Course catalogs group programs into 4 main categories: Credit & Banking, PACS Operations, Dairy & Agro-Processing, and Digital Management.
- **Fitts's Law (Optimizing Touch Targets):**
 - On mobile interfaces, all primary actions (attendance check-in, quiz submissions, audio

playback controls) provide large touch targets ($\geq 48 \times 48dp$) located in the lower thumb-zone.

- **Jakob's Law (Familiar Design Conventions):**

- The ERP timetable uses standard calendar paradigms familiar to administrative staff.
- E-learning course modules follow established LMS patterns (left-hand curriculum outline, center content viewport, bottom navigation buttons).

- **Error Prevention and Graceful Recovery:**

- Critical actions (such as rejecting nominations or reassigning hostel blocks) use double-confirmation dialogs with clear explanations.
- Destructive actions include a 10-second "Undo" toast notification at the bottom of the screen.
- Form inputs perform real-time inline validation, displaying clear, helpful error messages alongside input fields.

7. Module-by-Module UI/UX Layout Specifications

7.1 Online Programme Nomination & Batch Management

- **Administrative Nomination Table:**

- *Header Controls:* Search bar, filter dropdowns (State, District, Sponsoring Society Type), bulk action buttons ("Approve Selected", "Export CSV").
- *Data Table:* Columns for Trainee Name, Society Name, District, Verification Status, Prerequisite Check, and Action buttons.
- *Status Badges:* Clear visual indicators for Nominated (Blue), Verified (Amber), Batch Assigned (Emerald), and Conflict (Red).

- **Conflict Resolution Alert:**

- If a trainee is nominated for two overlapping courses, the system flags the entry with a subtle amber banner explaining the overlap and offering one-click batch reassignment.

7.2 Interactive Timetable & Hostel Logistics Management

- **Visual Timetable Grid:**

- Multi-resource calendar view showing classrooms and computer labs along the Y-axis and hourly time slots (09:00 to 18:00) along the X-axis.
- Drag-and-drop course blocks colored by subject category, with instant visual warnings if an instructor or room is double-booked.

- **Interactive Hostel Floor Map:**

- Visual floor plans of residential hostel blocks.
- Color-coded room tiles: Empty Bed (Neutral Slate), Occupied Bed (Coop Emerald), Reserved for Incoming Batch (Warm Amber), Maintenance (Striped Red).
- Clicking a room opens an occupancy side-panel displaying trainee names, sponsoring society details, check-in dates, and special requirements.

7.3 Digital Attendance Experience (Dual-Mode Interface)

- **Instructor Presentation Display (Dynamic QR):**

- A full-screen classroom presentation mode displaying the institute name, batch title, date, and a large centered QR code.

- A circular progress ring around the code counts down 15 seconds until the next automatic refresh.
- A real-time attendance counter at the bottom updates live as students scan in (e.g., "34 / 40 Students Present").
- **Trainee Mobile Scanner:**
 - Launches the device camera within a clear square targeting frame.
 - Instant haptic feedback and an animated green checkmark confirm when attendance is verified, displaying: "Attendance Recorded • Session 3: Common Accounting System".
- **Edge Biometric Kiosk Display:**
 - A clean ambient interface displaying: "Please look at the camera to check in".
 - Real-time face detection highlights the user's face with a smooth bounding box.
 - Upon verification, the screen displays a green confirmation card with the trainee's photo, name, and assigned seat for 1.5 seconds before resetting for the next person.

7.4 Offline-First Multilingual LMS & Assessment

- **Mobile Learning Viewport:**
 - *Header Area:* Displays the lesson title, language selector dropdown, and an offline download button with a progress ring.
 - *Audio/Video Player:* Includes adjustable playback speeds (0.75x, 1x, 1.25x), toggleable multilingual subtitles, and an Indic voice narration button.
 - *Text Content:* Clean typography with adjustable font sizing (A-, A, A+) and high-contrast color modes.
 - *Bottom Bar:* "Previous Lesson" and "Complete & Next" buttons anchored within easy reach of the user's thumb.
- **Assessment Interface:**
 - Clear progress indicator showing question progress (e.g., "Question 4 of 10").
 - Large, selectable answer choice cards with comfortable padding.
 - At the end of the quiz, a summary screen displays the score, pass/fail status, and direct links to review incorrect answers.

7.5 Verifiable Credentials & DigiLocker Export

- **Digital Credential Showcase Card:**
 - Modeled after an official certificate with a decorative border, the Ministry of Cooperation emblem, and the NCCT watermark.
 - Displays trainee name, course title, completion date, assessment grade, and issuing institute.
 - Includes an interactive, scannable QR code linking to the verification registry.
 - Features a prominent button: "**Add to DigiLocker**", which sends the certificate to the citizen's government wallet via API Setu.

7.6 Indic Conversational Career Counseling Drawer

- **Chat Interface Layout:**
 - A slide-over panel on desktop or full-screen view on mobile, featuring the assistant avatar ("सहकार मित्र / Sahkar Mitra").
 - Quick-prompt suggestion pills above the input field (e.g., "How do I become a PACS

Secretary?", "Courses for Dairy Management", "Jobs in my district").

- Messages display clear references to source policies, showing users where guidance originates.
- A large microphone button at the bottom allows users to speak their questions in regional languages via Bhashini voice integration.

7.7 Cooperative Employment Exchange & Matching Radar

- **Recruiter Dashboard:**
 - Clean overview cards tracking Active Jobs, Total Applicants, Verified Matches, and Scheduled Interviews.
 - A candidate list showing match scores, verified certificates, and geographic proximity.
- **Candidate Detail Modal:**
 - An interactive visual radar showing alignment across four dimensions: Verified Skills, NCCT Certifications, Practical Experience, and Geographic Proximity.
 - Displays verified certifications with green checkmarks, giving employers immediate confidence in candidate qualifications.

8. Impressive Features and Delightful

Micro-Interactions

- **Floating Indic Voice Assistant ("सहकार वाणी / Sahkar Vani"):**
 - A floating circular button anchored in the bottom right corner of the screen.
 - Tapping the button opens an animated voice-wave visualizer that listens to questions in any of the 22 Scheduled Languages, speaks the response aloud, and navigates to relevant course materials.
- **Offline Resilience Indicator ("Sahkar Suraksha Pill"):**
 - When internet connectivity is lost, the network indicator changes smoothly from green to slate: Offline Mode Active • All Work Saved Locally.
 - When the connection returns, a subtle green progress animation confirms: Synchronized 14 records with Central Portal.
- **Celebratory Course Completion Interaction:**
 - Upon passing an assessment and meeting attendance criteria, the system displays an animated badge unlock with light confetti effects.
 - The interface smoothly generates the verified digital credential and offers one-tap export to DigiLocker, creating a rewarding milestone for rural learners.
- **Interactive District Career Radar:**
 - A visual radar chart in the trainee portal that compares the learner's certified skills against real hiring demands from local PACS and cooperative societies in their home district, highlighting clear paths to employment.

9. Accessibility, Inclusive UX, and Vernacular Usability Matrix

Accessibility Vector	Implementation Standard & Mechanism	Target User Impact
Low-Literacy Voice Navigation	Integration with MeitY Bhashini ASR and TTS engines; voice-prompt buttons beside major text fields.	Enables rural trainees and SHG members with limited reading literacy to listen to course materials and use the platform.
High Contrast & Sunlight Legibility	High-contrast viewing mode (contrast-ratio $\geq 7:1$) with strong borders and saturated text tones.	Ensures screen readability on affordable mobile devices under bright outdoor sunlight.
Font Scaling & Ergonomics	User-controlled typography toggle (100%, 115%, 130%) without breaking responsive layouts.	Supports senior cooperative committee members and visually impaired users.
Bandwidth Optimization	Low-bandwidth mode that defaults to compressed illustrations and audio summaries before loading video files.	Reduces data consumption and prevents page load failures over 2G/3G rural networks.
Tactile & Audio Feedback	Short haptic vibrations and audio chimes on successful QR scans, quiz submissions, and sync events.	Provides immediate confirmation of completed actions for first-time digital platform users.

This UI/UX specification delivers an inclusive, enterprise-ready design framework that balances administrative functionality with accessible learning experiences, modernizing capacity building across India's cooperative sector.

Would you like me to do a more in-depth research instead and propose a research plan?